

User Interface On-boarding Tasks

General Prerequisites (Applies for all sub-teams):

- Completed [Setting Up Your Development Environment](#)
- Have a working knowledge of C++ and ROS2

Sub-Team Prerequisites:

- Completed the optional [Setting Up User Interface Workspace](#) instructions
- Have an introductory knowledge of HTML, CSS, and JS

Instructions: Students wishing to be on-boarded to the team should solve 1 problem. This is NOT a meaningless interview-style task. Instead, these problems are actual problems the sub-team is struggling with. This is your first real task on the sub-team. Students should refrain from completing multiple problems without prior approval to ensure all students wishing to be on-boarded have a task to complete. **Upon completion of problem, please meet with sub-team lead prior to pushing code changes.**

Fixing Pseudo-Terminal Output

Problem 1: Some topic logs (notably /GPSData) contain so much information that messages are cluttered and hard to separate from each-other.

- Languages Required: HTML, CSS, JS
- Frameworks Required: Roslib JS
- Suggested Approaches: adding spacing between messages, changing terminal dimensions, styling important text

Problem 2: Some topic logs (notably /base_station_heartbeat) overflow their container (making the terminal scrollable) before clearing it. The intended functionality is to clear the terminal as soon as the terminal is full; the terminal should never be scroll-able.

- Languages Required: HTML, CSS, JS
- Frameworks Required: Roslib JS
- Suggested Approaches: understand the current approach in order to identify and correct its flaws

Improving the GPS Map

Problem 1: When studying specific areas, the small dimensions of the map can be frustrating.

- Languages Required: HTML, CSS, JS
- Frameworks Required: None strictly required, Leaflet and Roslib JS could be beneficial
- Suggested Approaches: Add a full-screen toggle, implement dynamic scaling

Problem 2: While the map auto-focusing on the rover's position is useful for mid-task tracking, it can become frustrating when trying to scout other areas.

- Languages Required: HTML, CSS, JS
- Frameworks Required: Leaflet; Roslib JS could be beneficial
- Suggested Approaches: Add an auto-focus toggle